Transfer task: plan and begin

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Learning target: Organize the given evidence and begin an independent position prediction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 18.*
12 N left; 15-3-12=0 N.

02. Worked example: explain the model
A 10 m rightward path; later timed position observations can test equal motion per equal time.

03. Organize the givens
Correct table and sketch; initial x negative, velocity components positive; time in seconds, velocity km/s.

04. Find the central end point
Δx 131,360,000 km; xf 11,360,000 km;Δy=yf 35,200,000 km.

05. Start the bounds
Lower x = -120,050,000 + 8.02(16,000,000) = 8,270,000 km. Upper x = -119,950,000 + 8.40(16,000,000) = 14,450,000 km. y ranges from 2.15(16,000,000) = 34,400,000 km to 2.25(16,000,000) = 36,000,000 km.

06. Review: connect an earlier idea *Review: Unit 1, Day 9.*
The straight-moving rover can have zero net force. The turning rover changes velocity direction and needs net force.

07. Checkpoint
Assumption constant velocity; gravity/Earth motion omitted. Require visible independent work and specific next step.

08. STRETCH: test the model
Example: initial speed was lower, or object slowed; multiple timed positions distinguish initial velocity and acceleration.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 18.*
At rest T=29.4 N. Accelerating: T=m(g+a)=35.4 N.

02. Worked example: explain the model
A 10 m rightward path; later timed position observations can test equal motion per equal time.

03. Plan and represent
Axes fixed,+x initially Earthward,+y perpendicular; asteroid negative x. Plan must start consistent units.

04. Central prediction
v=(8.2103695, 2.1999619) km/s; xf 11,365,912.4 km; yf 35,199,390.1 km. Displacement=(131,365,912.4, 35,199,390.1) km.

05. Propagate bounded inputs
δx=50,000+(0.2 cos 15)(16,000,000)=3,140,962.6 km;δy=(0.2 sin 15)(16,000,000)=828,220.9 km. x 8,224,949.7 to 14,506,875.0; y 34,371,169.2 to 36,027,611.1 km.

06. Review: connect an earlier idea *Review: Unit 1, Day 9.*
Tangent velocity direction changes, so acceleration and net force are inward and nonzero. Constant speed is insufficient for equilibrium.

07. Checkpoint
Start: central -120 million km; bounds [-120.05, -119.95] million km. Final central x = 11.365912 million km; bounds [8.224950, 14.506875] million km. State constant velocity and an omitted effect such as gravity or Earth's motion.

08. STRETCH: test the model
v=1.6 m/s; or a=2×8/25=0.64 m/s² from rest. Intermediate positions distinguish the models; original 10 m is outside the stated bound.